XRPL Validator — Daily Engineering Report · 2026-04-10

Period: 2026-04-10 00:00 to 24:00 US Eastern (2026-04-10 04:00 to 2026-04-11 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 6 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-04-10): 0.99893 (23 missed of 21,433). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-node): correct terPRE_SEQ sender extraction + shadow hash off-by-1** (`dcc1d8e`, xrpl-validator) Two diagnosis-driven fixes from a 63K-ledger battle test that exposed 557 FFI divergences and 3 shadow-hash mismatches: ffi_engine.rs — terPRE_SEQ recovery was extracting the wrong account. The scan looked for any 0x81 byte and read the next 20 bytes, but (a) random 0x81 bytes appear inside SigningPubKey/TxnSignature blobs which canonically come BEFORE sfAccount, and (b) the slice was off by one — it included the VL …
- **feat(ui): cyberpunk PCB consensus board + sparkline transaction flow** (`db09a15`, xrpl-validator) Two main visualization upgrades on the validator dashboard: Transaction Flow — replaced the static bar chart with a sparkline-style stacked area: 900 data points, 20ms update tick, smooth left-to-right fill with a glowing dot at the rightmost active sample. Top-4 tx types labeled inline. Persists to localStorage so the chart survives refreshes. Pattern lifted from the metrics.html sparkline and tuned for fluid …
- **feat(xrpl-node): stage3 prep — config, batch helpers, hybrid singleton model** (`9d9f694`, xrpl-validator) Adds the building blocks for Stage 3 (FFI bytes as the source of truth in state.rocks) WITHOUT wiring them into the live loop. The wiring is deliberately deferred until after Phase A — the 100K -> 200K -> 300K shadow-only battle test currently running on the validator host — clears with a clean shadow hash record. Flipping Stage 3 with even a small shadow mismatch rate would silently corrupt state.rocks for the divergent ledgers.
- **fix(bulk_sync): post-build root verification + seed-in-retention check** (`0f8dcd7`, xrpl-validator) Investigation of the 91 missing SLEs from the prior battle test pointed at a class of silent-gap bugs in the bulk download path. None of the bugs were in our code per se — they're rippled-side partial responses that we trusted blindly: *

`ledger_data` returned partial pages when the seed ledger had any leaves rippled hadn't fully cached locally * a worker hitting >50 errors silently gave up mid-walk * the SHAMap …

- **refactor(xrpl-node): centralize data paths in xrpl_node::paths** (`2d8ecca` , xrpl-validator) The data-directory layout was split-brain across the codebase. `bulk_sync` read `XRPL_DATA_DIR` (with a buggy default that included `/sync`), while `live_viewer` hardcoded `[local path]` in nine places. The split was masked on the validator host by a `[local path] → ~/xrpl-data` symlink, so nothing visibly broke — but a fresh deployment without the symlink would have had the writer and the reader pointing at …

Tests and campaigns

- **test(xrpl-node): pin terPRE_SEQ extraction + shadow overlay against regressions** (`4f384a5` , xrpl-validator)

Independence program

No independence record for UTC day 2026-04-10; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
04-10 23:26	—	xrpl-validator	`dcc1d8e`	fix(xrpl-node): correct terPRE_SEQ sender extraction + shadow hash off-by-1
04-10 23:27	—	xrpl-validator	`db09a15`	feat(ui): cyberpunk PCB consensus board + sparkline transaction flow
04-10 23:45	—	xrpl-validator	`4f384a5`	test(xrpl-node): pin terPRE_SEQ extraction + shadow overlay against regressions

Committed	Merged into main	Repo	Commit	Subject
04-10 23:46	—	xrpl-validator	`9d9f694`	feat(xrpl-node): stage3 prep — config, batch helpers, hybrid singleton model
04-10 23:56	—	xrpl-validator	`0f8dcd7`	fix(bulk_sync): post-build root verification + seed-in-retention check
04-11 00:06	—	xrpl-validator	`2d8ecca`	refactor(xrpl-node): centralize data paths in xrpl_node::paths

Times in UTC. A commit appears on the day it was committed to main or merged into it.